One of the most effective treatments for alcoholism or drug addiction—indeed, much research suggests it is *the* most effective treatment—is family therapy (Stanton & Shadish, 1997). By improving the person's family life, possibly improving the job situation, and helping the person to develop new interests, the therapy decreases the person's compulsion for alcohol or drug use.

***The Controlled Drinking Controversy*** Most physicians agree with Alcoholics Anonymous that the only hope for an alcoholic is total abstinence. Drinking in moderation, they insist, is out of the question.

A few psychologists, however, are not convinced that abstinence is the best advice for *all* alcoholics (Peele, 1998). Some alcoholics who try to abstain repeatedly fail; a few of these people do learn to reduce their drinking without eliminating it altogether. This is not to say that alcoholics can simply decide to drink in moderation; if they could, they would not have become alcoholics in the first place. Rather, the point is that a few people who fail to stick to an abstention program can learn (with difficulty) to drink less than they have been, to stay out of legal trouble, and in general to do themselves less damage. Psychologists have established several programs to try to teach alcoholics "controlled drinking," with at least occasional success (Rosenberg, 1993). Critics charge that even offering the hope of controlled drinking decreases people's motivation to abstain. Defenders of controlled drinking reply that some people still find themselves unable to abstain despite their best efforts.

One of the biggest unresolved problems in psychotherapy is matching individuals to the best treatment. A similar issue arises with regard to weight loss, depression, substance abuse, and many other fields in which therapists offer several approaches, each of which appears to be effective for some people and not for others. When treating substance abuse, the options include AA and other abstention-based programs, controlled drinking programs, family therapy, Antabuse, and psychotherapy aimed at the user's other (nondrug-related) psychological disorders. Several attempts have been made to determine each person's needs and match each to the most appropriate treatment. So far, such procedures have produced results that are only slightly better than assigning people to therapies at random (McLellan et al., 1997). Perhaps future research will enable more effective matching.

## *Specific Treatments for Opiate Addiction*

Before the year 1900, opiate drugs such as morphine and heroin were considered far less dangerous than alcohol (Siegel, 1987). In fact, many medical doctors used to urge their alcoholic patients to switch from alcohol to morphine. Then, around 1900, the use of opiates was made illegal in the United States, except by prescription to control pain. Since then, research on opiate use has been limited by the fact that only lawbreakers now use opiates.

Some users of heroin and other opiates try to break their habit by going "cold turkey"—abstaining altogether until the withdrawal symptoms subside, sometimes under medical supervision. Many people, however, experience a recurring urge to take the drug, even long after the withdrawal symptoms have subsided. For those who cannot quit, researchers have sought to find a nonaddictive substitute that would satisfy the craving for opiates without creating harmful effects. Heroin was originally introduced as a substitute for morphine, before physicians discovered that heroin is even more addictive and troublesome than morphine!

Today, the drug **methadone** (METH-uh-don) *is commonly offered as a less dangerous substitute for opiates.* Methadone is chemically similar to both morphine and heroin and can itself be addictive. (Table 16.4 compares methadone and morphine.) When methadone is taken in pill form, however, it enters the bloodstream gradually and also departs gradually (Dole, 1980). (If morphine or heroin is taken as a pill, much of the drug is broken down in the digestive system and never reaches the brain.) Thus, methadone does not produce the "rush" associated with intravenous injections of opiates; nor does it produce rapid withdrawal symptoms. Although methadone satisfies the craving for opiates without seriously disrupting the user's behavior, it does not eliminate the addiction itself. If the dosage is reduced, the craving returns.

**TABLE 16.4 Comparison of Methadone with Morphine**

| | MORPHINE | METHADONE BY INJECTION | METHADONE TAKEN ORALLY |
|---|---|---|---|
| *Addictive?* | Yes | Yes | Weakly |
| *Onset* | Rapid | Rapid | Slow |
| *"Rush"?* | Yes | Yes | No |
| *Relieves craving?* | Yes | Yes | Yes |
| *Rapid withdrawal symptoms?* | Yes | Yes | No |

Going cold turkey: Heroin withdrawal resembles a severe bout of the flu, with aching limbs, intense chills, vomiting, and diarrhea; it lasts a week, on average. Unfortunately, even after people have suffered through withdrawal, they are likely to experience periods of craving for the drug.

Many addicts who stick to a methadone maintenance program hold down a job and commit fewer crimes than they did when they were using heroin or morphine (Woody & O'Brien, 1986). Some of them, after discovering that they can no longer get high from opiates, turn instead to a nonopiate drug such as cocaine (Kosten, Rounsaville, & Kleber, 1987). In other words, methadone maintenance programs do not eliminate addictive behaviors. At present, there is no reliable cure for opiate dependence.

## THE MESSAGE
## *Substances, the Individual, and Society*

Substance abuse is costly to society as well as to the individual. The United States and many other countries have tried to control the problem by issuing stiff prison sentences for those who use or sell certain drugs. The result? Hard to say; drug use continues to be widespread, but we do not know how much worse it might be without the legal prohibitions. However, we do know that law enforcement itself is costly; prisons are crowded with those whose only offense was drug related, and many drug users turn to theft or violence as a way of getting drugs. The Netherlands has experimented with greatly decreased penalties against the use and sale of drugs, especially marijuana. The results are in some ways favorable, in other ways unfavorable; overall, it is impossible to generalize about what might happen if other countries tried the same approach (MacCoun & Reuter, 1997). As a voter, you will have to help our society make decisions about how to deal with substance abuse, a problem that shows no signs of becoming less difficult.

## SUMMARY

* *Substance dependence.* People who find it difficult or impossible to stop using a substance are said to be dependent on it or addicted to it. (page 605)
* *Addictive substances.* Generally, the faster a substance enters the brain, the more likely it is to be addictive. Cigarette smokers inhale enough nicotine to satisfy their addiction, even if they smoke low-nicotine brands. (page 606)
* *Predisposition to alcoholism.* Some people may be predisposed to become alcoholics for genetic or other reasons. People at risk for alcoholism find that alcohol relieves their stress more than it does for other people. People who underestimate their level of intoxication are more likely than others to become alcoholics later. (page 606)
* *Alcoholics Anonymous.* The most common treatment for alcoholism in North America is provided by the self-help group called Alcoholics Anonymous. (page 609)
* *Antabuse.* Some alcoholics are treated with Antabuse, a prescription drug that makes them ill if they drink alcohol. (page 609)
* *The "disease" concept.* Whether or not substance abuse is considered a disease depends on what we mean by "disease." The long-term course varies among different alcoholics or drug abusers. One of the most effective treatments for substance abuse is family therapy, which takes a nonmedical approach. (page 610)
* *The "controlled drinking" controversy.* Whether certain alcoholics can be trained to drink in moderation is a controversial question. Therapists hope to improve their ability to assign each person to the most effective treatment. (page 611)
* *Treatments for opiate abuse.* Some opiate users quit using opiates, suffer through the withdrawal symptoms, and manage to abstain from further use. Others substitute methadone under medical supervision. Although methadone has fewer destructive effects than morphine or heroin, it does not eliminate the underlying dependence. (page 611)

## Suggestion for Further Reading

Marlatt, G. A., & VandenBos, G. R. (Eds.). (1997). *Addictive behaviors: Readings on etiology, prevention, and treatment.* Washington, DC: American Psychological Association. A collection of research articles on the nature of addiction and the methods of treatment.

## Terms

**dependence** or **addiction** a self-destructive habit that someone finds difficult or impossible to quit (page 605)

**alcoholism** habitual overuse of alcohol (page 607)

**Type I** (or Type A) **alcoholism** the type that is generally less severe, equally common in men and women, less dependent on genetics, and likely to develop gradually, presumably in response to the difficulties of life (page 607)

**Type II** (or Type B) **alcoholism** the type that is generally more severe, more common in men, more often associated with aggressive or antisocial behavior, more dependent on genetics, and likely to begin early in life (page 607)

**detoxification** a supervised period for removing drugs from the body (page 609)

**Alcoholics Anonymous (AA)** a self-help group of people who are trying to abstain from alcohol use and to help others do the same (page 609)

**Antabuse** the trade name for disulfiram, a drug used in the treatment of alcoholism (page 610)

**methadone** a drug commonly offered as a less dangerous substitute for opiates (page 611)

## Answers to Concept Checks

7. The injection route is more likely to lead to addiction. Other things being equal, the faster that a drug reaches the brain, the more likely it is to become addictive. (page 606)
8. They are less likely than others to become alcoholics. This gene is considered the probable reason that relatively few Asians become alcoholics (Harada, Agarwal, Goedde, Tagaki, & Ishikawa, 1982; Reed, 1985). (page 610)

## Web Resources

**Substance Abuse**

www.psych.org/public_info/substance.html

In *Let's Talk Facts* pamphlets, use and abuse is examined with regard to alcohol, marijuana, cocaine, opiates, hallucinogens, inhalants, sedative hypnotics, and nicotine.

MODULE 16.3

# Mood Disorders

Why do people become depressed?

What can be done to relieve depression?

Even when things are going badly, most people remain optimistic that all will be well in the end. After the hurt and disappointment we feel when a relationship breaks up, we say, "Oh, well, at least I learned something from the experience." When we lose money, we say, "It could have been worse. I still have my health."

But sometimes we feel depressed. Nothing seems as much fun as it used to be, and the future seems ominous. For some people, the depression is severe and long-lasting. Why?

## Depression

When people say "I'm depressed," they often mean "I am sad; life isn't going very well for me right now." In psychology, **major depression** refers to a much more *extreme condition, persisting most of each day for a period of months, in which the person experiences little interest in anything, little pleasure, little reason for any productive activity.* Aaron Beck (1973) described one depressed woman who stood in front of an elevator for 15 minutes because she did not have enough desire to press the button.

Sad people are unhappy at the moment, whereas depressed people cannot even imagine anything that would make them happy. They have trouble concentrating. Their appetite and interest in sex decrease (Nofzinger et al., 1993). They feel worthless, fearful, guilty, and powerless to control what happens to them. Most severely depressed people consider suicide, and many attempt it. As with any other psychological disorder, depression varies in severity from one person to another and from one time to another (Flett, Vredenburg, & Krames, 1997).

Nearly all depressed people experience sleep abnormalities (Carroll, 1980; Healy & Williams, 1988). (See Figure 16.14.) They enter REM sleep less than 45 minutes after falling asleep (an unusually short time for most people). Most depressed people wake up too early and cannot get back to sleep. When morning comes, they feel poorly rested. In fact, they usually feel most depressed early in the morning. During most of the day, they feel a little sleepy.

Depression is common from adolescence through old age; the peak time for first diagnosis is at age 30–40. Regardless of the age at diagnosis, most people say they had been somewhat depressed for years before the condition became severe enough for them to visit a therapist (Eaton et al., 1997). Depression occurs in episodes; typically, a person might feel depressed for a few months, then recover for months or even years, and then experience the next episode of depression. In rare cases, an episode can last for years at a time. On the average, earlier episodes last a bit longer than later episodes (Solomon et al., 1997).

**Bipolar disorder,** previously known as *manic-depressive disorder,* is a related condition in which *a person alternates between periods of depression and periods of mania, which is the opposite extreme.* We shall return to bipolar disorder later.

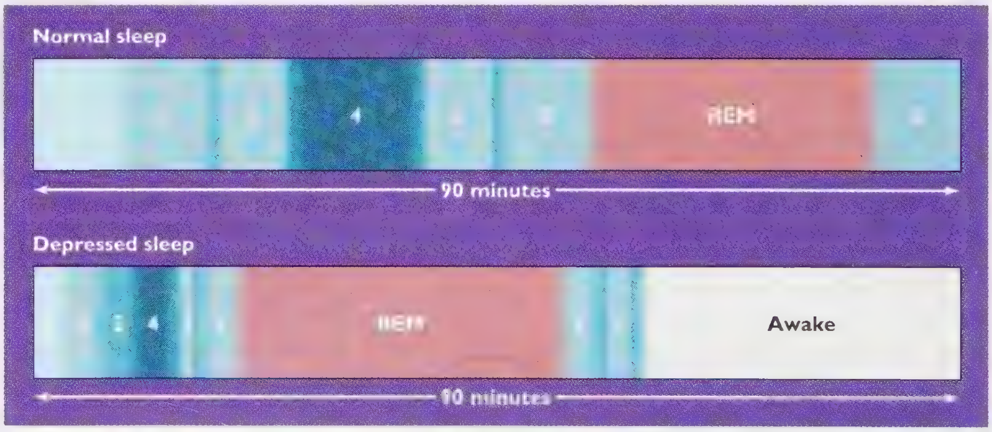


**FIGURE 16.14** When most people go to sleep at their usual time of day, they progress slowly to stage 4 and then back through stages 3 and 2, reaching REM sleep toward the end of their first 90-minute cycle. Depressed people, however, reach REM more rapidly, generally in less than 45 minutes. They also tend to awaken frequently during the night.

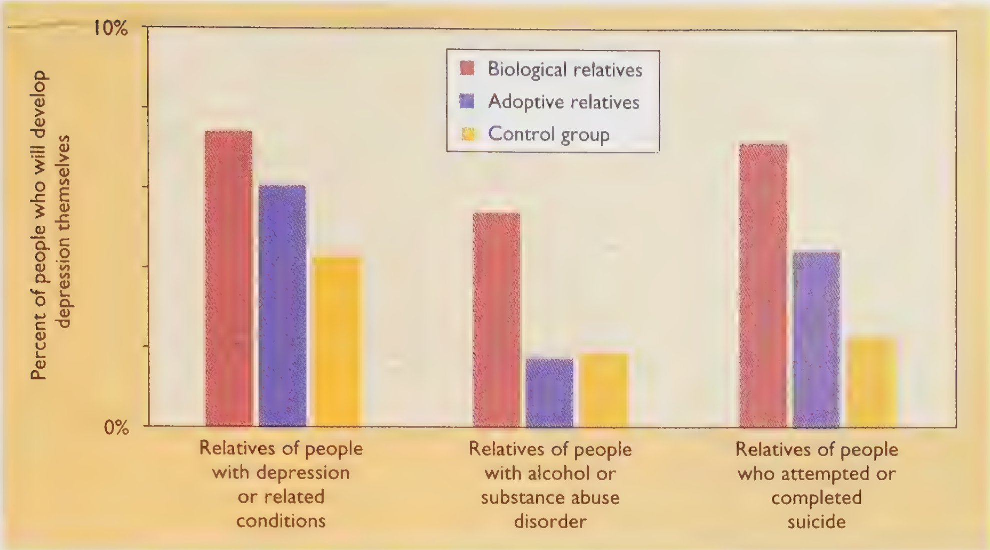


**FIGURE 16.15** Blood relatives of a depressed person are more likely to suffer depression themselves, when compared with incidences of depression in the general public, which is 5–10%. The closer the genetic relationship is, the greater is the probability of experiencing depression.

## Genetic Predisposition to Depression

The fact that depression tends to run in families suggests that some people have a genetic predisposition to depression (see Figure 16.15). Children and other relatives of depressed people are more likely than other people to become depressed themselves in adulthood and also somewhat more likely than others to develop phobias, panic disorder, substance abuse, and social impairments (Kendler, Heath, Neale, Kessler, & Eaves, 1993; Kendler, Neale, Kessler, Heath, & Eaves, 1992; Weissman, Warner, Wickramaratne, Moreau, & Olfson, 1997). It is likely that certain families have genes that express themselves in different ways, leading to depression in some people and to other disorders in others.

Adopted children who become depressed usually have more biological relatives than adopting relatives who are depressed (Wender et al., 1986). So far, we do not have consistent evidence to indicate whether depression depends on one gene or several (Faraone, Kremen, & Tsuang, 1990). Chances are that a disposition to depression depends on different genes or gene combinations in different families.

## The Sex Difference in the Prevalence of Depression

From adolescence onward, women are about twice as likely as men to experience major depression and as much as four times as likely to experience severe depression. This ratio varies among cultures, but the tendency is in the same direction for North America, Europe, New Zealand, and Korea (Culbertson, 1997).

Why is depression more common in women than it is in men? One possibility is that hormonal changes in women's menstrual cycles or occurring as a result of pregnancy and childbirth trigger episodes of depression. For example, *shortly after giving birth,* a time of massive hormonal changes, some women enter a **postpartum depression.** Estimates of the frequency of postpartum depression vary widely, depending on whether one counts only the severe cases (about 1 per 1,000), the moderate cases (about 1 in 10), or the mild cases (about 3 in 10). Most of the women who suffer moderate to severe postpartum depression, however, have a history of other depressive episodes; the hormonal swing after giving birth is not so much a cause of depression as a trigger for an additional depressive episode (O'Hara, Schlechte, Lewis, & Wright, 1991).

The evidence is not convincing that hormones account for the higher prevalence of depression in women than in men. Psychologists have considered several other

Depression is most common among people who have little social support.

hypotheses: that women are more likely than men to report their depression and to seek help, that many depressed men receive a diagnosis of alcoholism instead of depression, and that women become depressed in reaction to their lower status and lack of power. Although each of these possibilities probably contributes to the sex difference in depression, the evidence does not strongly support any of them.

Susan Nolen-Hoeksema (1990, 1991) has suggested another possibility: When men start to feel depressed, they generally try to distract themselves. They try not to think about whatever is making them depressed; they play basketball or watch a movie or do something else that they enjoy. Women are more likely to ruminate—to think about why they are depressed, to talk with others about their feelings, even to have a long cry. According to Nolen-Hoeksema, ruminating about depression only makes it worse. The ruminative thoughts interfere with useful problem solving and bias a person toward a pessimistic appraisal of the situation. This explanation has the advantage of suggesting ways to help women (and men) avoid or minimize their depression. It does not, however, address the question of *why* women ruminate more and distract themselves less than men do.

### Events That Precipitate Depression

As a rule, people become depressed when bad things happen to them. For example, most people are clearly depressed for at least the first two months after the death of a spouse (Thompson, Gallagher-Thompson, Futterman, Gilewski, & Peterson, 1991). However, the severity of an unpleasant event is a poor predictor of how depressed a person will become following that event. Some people become depressed without any apparent stressful experience (Brown, Harris, & Hepworth, 1994), and some depressed people bring about their own stressful events (Hammen, 1991).

One study illustrated the differences among people in their responses to a stressful event by taking advantage of the fact that students in one California university had answered questionnaires about their mental health 2 weeks before a major earthquake in 1989. Psychologists asked the same students to fill out questionnaires again 10 days and 7 weeks after the earthquake. The result was that students who were already somewhat depressed before the earthquake became very depressed afterward; students who were not depressed earlier suffered some distress but in most cases recovered rapidly (Nolen-Hoeksema & Morrow, 1991). In other words, people who are vulnerable to depression are more likely to react strongly to any stressful event.

So, what makes some people more vulnerable than others? One explanation is that severe losses early in life make people more vulnerable to depression later on. For example, adolescents who lose a parent through death or divorce are likely to react strongly to other losses later in life, even to routine events such as breaking up with a boyfriend or girlfriend (Roy, 1985).

People with poor social support also tend to be vulnerable to depression. As we saw in Chapter 12, social support helps people to cope with stress. People who are in a happy marriage and who have close friends are less likely to become depressed or to remain depressed than people who have no one to talk to about their troubles, or people who fail to make use of the emotional support that their friends offer them (Rivera, Rose, Futterman, Lovett, & Gallagher-Thompson, 1991).

What most influences the onset of depression is not what events transpire but how people interpret them. For example, a trivial event such as not being invited to a party might contribute toward depression for someone who regarded the noninvitation as evidence of rejection by other people (Johnson & Roberts, 1995). To understand who becomes depressed and why, we need to understand how people think.

## Cognitive Aspects of Depression

Most people believe that every cloud has a silver lining. Show depressed people a silver lining, though, and they wrap it in a cloud. Somehow, they think differently from people who are not depressed. Do their thought patterns lead to their depression?

### The Learned-Helplessness Model

According to the **learned-helplessness theory**, *people become depressed if their experiences teach them that they have no control over the major events in their lives.* The learned-helplessness theory originated while testing theories about avoidance learning in animals. Steven Maier, Martin Seligman, and Richard Solomon (1969) strapped some dogs with no previous training into restraining harnesses and then repeatedly sounded a tone that was followed by a shock to the dogs' feet. The dogs soon learned that the tone predicted shock. They struggled to escape but could not. The next day, the experimenters put the same dogs into an apparatus in which a tone served as a warning signal for an escapable shock. Ordinarily, dogs quickly learn a response that enables them to escape or avoid a shock. However, the dogs that had previously received inescapable shocks were extremely slow to learn, and most never did learn the avoidance response.

Why? The dogs had learned on the first day that the tone predicted shock; they had also learned that the shock was inescapable. When they started receiving shocks on the second day, they thus made no effort to escape. They had learned that they were helpless.

These "helpless" dogs resembled depressed people in several respects. The dogs were inactive and slow to learn. Even their posture and "facial expressions" suggested sadness. The experimenters proposed that the same process might operate in humans: People who meet with only defeat and loss, despite their best efforts, may come to feel helpless and therefore fall into depression.

## *Explanatory Styles*

The learned-helplessness hypothesis, though appealing, is no longer considered a viable explanation for depression, at least not in its original form. Experiments with humans failed to confirm that struggling with impossible tasks makes people feel depressed. It is not *failure* itself that makes people depressed, but *why people think they have failed.*

**TRY IT YOURSELF**

For example, suppose you fail a test. Which kind of explanation would you be likely to offer?

- The test was extremely difficult. Most other students probably did badly too.
- I had a weaker background from previous courses than most other students.
- I was sick and didn't get a chance to study.
- I'm just stupid. I always do badly, no matter how hard I try.

Using any of the first three attributions for failure, you probably wouldn't feel very depressed. You would be attributing your failure to a temporary, specific, or correctable situation—a problem that has nothing to do with your abilities. But the fourth attribution applies to you at all times in all situations. If you make that attribution—and if your grades are important to you—you are likely to feel depressed about your failure (Abramson, Seligman, & Teasdale, 1978; Peterson, Bettes, & Seligman, 1985). If you continue making similar attributions in many other situations, your depression may grow.

In a given situation, such as a low grade on a test, you might have a good reason for one attribution or another. For example, perhaps you really are the only one in this French class who did not take high-school French. Still, everyone has an **explanatory style,** *a tendency to accept one kind of explanation for success or failure more often than others.* Recall from Chapter 14 that an *internal attribution* cites a cause within the person. ("I failed the French test because I did not study properly. I did well on the biology test because I spent many hours studying.") An *external attribution* identifies a causes outside the person. ("I failed the French test because it was extremely hard. I did well on the biology test because it was easy.") People are not always consistent about how they explain their successes, but they tend to be very consistent, even over decades, about how they explain their failures. That is, people who generally blame themselves for their failures today will probably still be doing so 30 or 40 years from now (Burns & Seligman, 1989).

People who consistently take the blame for their own failures are said to have a *pessimistic* explanatory style, especially if their explanations for failure are stable and global. For example, "I failed the test because I did not study hard" is an unstable explanation, because you can study harder next time. "I failed the test because I'm not good at learning foreign languages" is a stable explanation, because it implies a permanent characteristic. "I failed the test because I'm stupid" is not only stable but also global; it applies not only to foreign languages but to all kinds of learning.

Researchers can determine people's explanatory style by asking them to explain some of their successes and failures. They can also gauge the explanatory styles of famous people, even dead people, by reading their speeches and writings to find out what explanations they offered for successes and failures. Using this method, psychologists have found that pessimistic political leaders tend to be cautious, indecisive, and inactive. Leaders with an optimistic explanatory style (the opposite of a pessimistic style) tend to be bold, active, even risk-takers (Satterfield & Seligman, 1994; Zullow, Oettingen, Peterson, & Seligman, 1988). (See Figure 16.16.) Athletes with a pessimistic style tend to give up after a defeat; for them, one defeat leads to another.

**FIGURE 16.16** During the Gulf War in 1991, the speeches of U.S. President George Bush and Iraqi leader Saddam Hussein varied between an optimistic and a pessimistic style from one period of time to another. Both leaders tended to make bold, risky decisions during periods when their speeches were optimistic; both were passive and cautious during periods when their speeches were pessimistic (Satterfield & Seligman, 1994).

Athletes with a more optimistic style keep on trying and even try harder after a defeat because, after all, they believe that the defeat was not their fault (Seligman, Nolen-Hoeksema, Thornton, & Thornton, 1990).

Although pessimism is hardly the same thing as depression, people with a pessimistic style are more likely than others to become depressed. Indeed, according to Aaron Beck (1973, 1987), depressed people consistently put unfavorable interpretations on the events of their lives. If someone walks by without comment, their likely response is, "See, people ignore me. They don't like me." After any kind of loss or defeat, "I'm a loser. I'm hopeless." After any kind of win, "That was just luck. I had nothing to do with it." Depressed people exaggerate their failures, minimize their successes, and give themselves a very low evaluation. Given this explanatory style, almost any event seems further grounds for feeling depressed.

### CONCEPT CHECK

**9.** Would depressed people be more or less likely than nondepressed people to buy a lottery ticket? (Check your answers on page 625.)

## *Therapies for Depression*

Depression is often a severe, even incapacitating disorder, but it often responds well to treatment. Both psychotherapy and drug therapy are effective.

### *Cognitive Therapy*

According to Aaron Beck, a pioneer in cognitive therapy, depressed people are guided by certain thoughts or assumptions of which they are only dimly aware. He refers to the "negative cognitive triad of depression":

- I am deprived or defeated.
- The world is full of obstacles.
- The future is devoid of hope.

Based on these assumptions, which Beck calls "automatic thoughts," depressed people interpret ambiguous situations to their own disadvantage. When something goes wrong, they blame themselves: "I'm worthless, and I can't do anything right." When an acquaintance walks past without smiling, they think, "She doesn't like me." They do not even look for alternative interpretations of a situation (Beck, 1991). Research has confirmed that depressed people are more likely than other people to agree with such statements as "I'm a loser," or "nothing ever works out right for me, and it's all my fault," or "I never have a good time" (Hollon, Kendall, & Lumry, 1986).

The task of a cognitive therapist is to help depressed people to substitute more favorable beliefs. Unlike rational-emotive therapists, who in many cases simply tell their clients what to think, cognitive therapists try to get their clients to make discoveries for themselves. The therapist focuses on one of the client's beliefs, such as "no one likes me." The therapist points out that this is a hypothesis, not an established fact, and invites the client to test the hypothesis as a scientist would: "What evidence do you have for this hypothesis?"

"Well," a client may reply, "when I arrive at work in the morning, hardly anyone says hello."

"Is there any other way of looking at that evidence?"

"Hmm. . . . I suppose it's possible that the others are busy."

"Does anyone ever seem happy to see you?"

"Well, maybe. I'm not sure."

"Then let's find out. For the next week, keep a notebook with you and record every time that anyone smiles or seems happy to see you. The next time I see you, we'll discuss what you've discovered."

The therapist's goal is to encourage depressed clients to discover that their automatic thoughts are incorrect, that things are not nearly as bad as they seem, and that the future is not hopeless. If one of the client's thoughts does turn out to be accurate—for example, "my boyfriend is interested in someone else"—then, the therapist asks, "Even if it's true, is that the end of the world?"

### *Biological Therapies for Depression*

Three types of drugs are currently used as antidepressants: tricyclics, monoamine oxidase inhibitors, and second-generation antidepressants. **Tricyclic drugs** (such as imipramine, trade name Tofranil) *block the reabsorption of several neurotransmitters—dopamine, norepinephrine, and serotonin—after they are released by an axon's terminal* (Figure 16.17). Thus, tricyclics prolong the effect of these neurotransmitters on the receptors of the postsynaptic cell. **Monoamine** (MAHN-oh-ah-MEEN) **oxidase inhibitors (MAOIs)** (such as phenelzine, trade name Nardil) *block the metabolic breakdown of released dopamine, norepinephrine, and serotonin* (Figure 16.17c). Thus, MAOIs also prolong the ability of released neurotransmitters to stimulate the postsynaptic cell. Tricyclic drugs are more effective than MAOIs for most patients; the MAOIs are used mainly when people fail to respond to tricyclics (Thase, Trivedi, & Rush, 1995). **Second-generation antidepressants** (such as fluoxetine, trade name Prozac) are also known as *serotonin reuptake blockers*. Like tricyclic drugs, *they act by blocking the reuptake of released neurotransmitters, but their effect is more narrowly limited to the neurotransmitter serotonin.* One advantage of the second-generation antidepressants is that they produce fewer side effects with most people; consequently, most people can take them in larger dosages and experience greater relief from depression (Burrows, McIntyre, Judd, & Norman, 1988).

The effects of antidepressant drugs build up gradually. Some depressed people report feeling better within the first

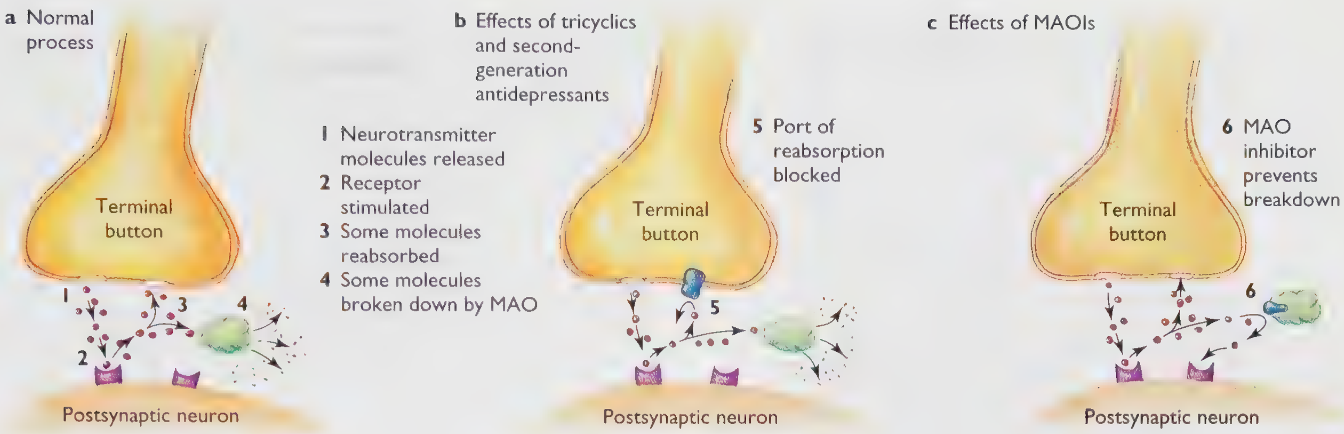


**FIGURE 16.17** Antidepressant drugs prolong the activity of the neurotransmitters dopamine, norepinephrine, and serotonin. (a) Ordinarily, after the release of one of the neurotransmitters, some of the molecules are reabsorbed by the terminal button, and other molecules are broken down by the enzyme MAO (monoamine oxidase). (b) Second-generation antidepressants prevent reabsorption of serotonin. Tricyclic drugs also prevent reabsorption but are less specific, acting on dopamine, norepinephrine, and serotonin. (c) MAO inhibitors block the enzyme MAO and thereby prolong the effects of the neurotransmitters.

week or two of use. The early benefit is almost always a *placebo effect*—that is, it depends on the person's expectation of feeling better and not on the drug itself. People who report quick benefits can be switched to an inactive substance such as sugar tablets without any loss of benefit (Stewart et al., 1998). The drug itself begins to show effects after 2 to 3 weeks of use, and the effects increase over the following 4–5 weeks (Blaine, Prien, & Levine, 1983).

What causes the gradual change in response? The biochemical effects are complicated. The drugs increase the quantity of neurotransmitter molecules bombarding the receptors fairly quickly, but they also initiate a slow series of changes in both the presynaptic and postsynaptic neurons, including decreases in receptor sensitivity, increases in neurotransmitter release, and increased production of certain chemicals that promote the growth and metabolism of neurons (Duman, Heninger, & Nestler, 1997; McNeal & Cimbolic, 1986). Many of these changes develop gradually over weeks. Researchers are still far from completely understanding how all these biochemical changes combine to produce antidepressant effects.

## CONCEPT CHECK

**10.** The drug mianserin prolongs the release of dopamine, norepinephrine, and serotonin from the terminal button. Would mianserin increase or decrease the intensity of depression? (Check your answer on page 625.)

## *The Advantages and Disadvantages of Antidepressant Drugs*

Overall, cognitive therapy helps a slightly higher percentage of depressed patients than drug therapy does, and its benefits generally persist longer after the end of therapy (Robinson, Berman, & Neimeyer, 1990). Cognitive therapy also has the advantage of producing no side effects, whereas about one-third of people taking tricyclic drugs report dry mouth, dizziness, sweating, or constipation (Blaine et al., 1983). Nevertheless, from the mid-1980s to the mid-1990s, the use of antidepressant drugs increased greatly, especially for second-generation antidepressants such as Prozac (Olfson et al., 1998). You might wonder why.

The reasons for this are convenience and lower cost. A client in psychotherapy will visit a therapist for at least 1 hour per week. Most therapists and clients agree that the benefits increase slowly over time, so the recommended course of treatment will last at least a couple of months, probably longer. The cost varies, depending on the therapist and the geographical region, but let's use $100 per hour as an approximation. If you are the client and you have enough available time and either have enough money yourself or have a generous insurance program that will pay for many visits, then psychotherapy is a good idea. However, if you are short of either time or money, you might be tempted by some relatively inexpensive pills. Antidepressants are often prescribed for people with anxiety disorders, obsessive-compulsive disorder, and adjustment problems, as well as depression.

Double-blind studies have consistently found that 50–70% of the adults who take tricyclic drugs experience improvements in their moods, as compared to 20–30% of those who take placebos (Blaine, Prien, & Levine, 1983; Gerson, Plotkin, & Jarvik, 1988; Morris & Beck, 1974). Similarly, talk-based psychotherapy produces benefits for most clients, but not all. Do drugs help the same people who would respond well to psychotherapy, or different people? Researchers don't know, and certainly no one knows how to identify which people might respond better to one treatment than the other.

For people with mild or moderate depression, a combination of both drugs and psychotherapy is not much more effective than either one alone—a result that suggests that drugs and psychotherapy are effective for mostly the same

**FIGURE 16.18** Electroconvulsive therapy is administered today only with the patient's informed consent. ECT is given in conjunction with muscle relaxants and anesthetics, to minimize discomfort.

people. However, for people with severe depression, a combination of drugs and psychotherapy can offer significant benefits (Thase et al., 1997).

## *Electroconvulsive Shock Therapy*

Another well-known but controversial treatment for depression is **electroconvulsive therapy,** abbreviated **ECT** (Figure 16.18): *A brief electrical shock is administered across the patient's head to induce a convulsion similar to epilepsy.* ECT was first used in the 1930s and became popular in the 1940s and 1950s as a treatment for schizophrenia, depression, and many other psychiatric disorders. It then fell out of favor, partly because antidepressant drugs and other therapeutic methods had become available and partly because ECT had been widely abused. Some patients were subjected to ECT hundreds of times, without their consent, and in many cases, ECT was used as a threat to enforce patients' cooperation.

Beginning in the 1970s, ECT has made a comeback in modified form, mostly for severely depressed people who fail to respond to antidepressant drugs, whose thinking is seriously disordered, or who have strong suicidal tendencies (Scovern & Kilmann, 1980). For suicidal patients, ECT has the advantage of taking effect rapidly, generally within 1 week. When a life is at stake, rapid relief is important.

ECT is now used only with patients who have given their informed consent. The shock is less intense than it used to be, and the patient is given muscle relaxants and anesthetics to prevent injury and to reduce discomfort. As a rule, the procedure is applied every other day for about 2 weeks; then, the physician evaluates the progress and either stops the treatment or decides to continue a few more times.

Exactly how ECT works is uncertain. Some have suggested that it relieves depression by causing people to forget certain depressing thoughts and memories. However, the data do not support that suggestion. Although ECT usually does impair memory, at least temporarily, there are ways to reduce the memory loss without lessening the antidepressant effect (McElhiney et al., 1995). ECT exerts a great many effects on the brain, including enhanced production of dopamine receptors (Smith, Lindefors, Hurd, & Sharp, 1995), decreased production of norepinephrine receptors (Kellar & Stockmeier, 1986), and increased release of both dopamine and norepinephrine (Chiodo & Antelman, 1980). At present, we do not know which of these effects is critical for the antidepressant outcome.

The use of ECT continues to be controversial. According to extensive reviews of the literature, ECT relieves depression with about 80% of the patients who receive it and generally produces fewer side effects than antidepressant drugs (Fink, 1985; Janicak et al., 1985; Weiner, 1984). However, ECT's benefits are generally temporary; about half of those who show a good response will relapse into depression within 6 months unless they receive some other therapy to prevent it (Riddle & Scott, 1995). Furthermore, the procedure's history of abuse has given it a bad reputation. Many psychiatrists therefore hesitate to recommend ECT.

## *Seasonal Affective Disorder*

One variety of depression is known as **seasonal affective disorder,** or **depression with a seasonal pattern** (Figure 16.19). People with this disorder *become seriously depressed once per year during a particular season.* Although

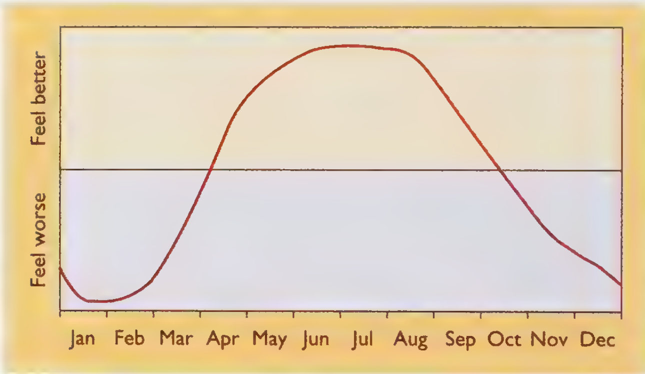


**FIGURE 16.19** Most people feel slightly better during the summer (when the sun is out most of the day) than during the winter (when there are fewer hours of sunlight). People with seasonal affective disorder feel good in the summer and seriously depressed in the winter (or good in the winter and depressed in the summer). Seasonal affective disorder is commonest in far-northern locations such as Scandinavia, where the summer days are very long and bright and the winter days are very short and dark. The disorder is unheard-of in tropical locations such as Hawaii, where the amount of sunlight per day varies only slightly between summer and winter.

annual winter depressions have received the most publicity, annual summer depressions also occur (Faedda et al., 1993). Unlike most other depressed patients, people with seasonal affective disorder tend to sleep and eat excessively during their depressed periods (Jacobsen, Sack, Wehr, Rogers, & Rosenthal, 1987).

People with the winter variety of seasonal affective disorder respond to the amount of sunlight they see each day. Most of us are more cheerful when the sun is shining than we are on cloudy days, but these people are unusually sensitive to the effects of sunlight. Seasonal affective disorder can be relieved by sitting for a few hours each day in front of a bright light. Exactly how the light relieves depression is still unknown.

## Bipolar Disorder

People with bipolar disorder (also known as manic-depressive disorder) alternate between the extremes of mania and depression. In most respects, **mania** is the *opposite of depression.* When people with bipolar disorder are in the depressed phase, they are slow, inactive, and inhibited. When they are in the manic phase, they *are constantly active and uninhibited.* When depressed, they feel helpless, guilt-ridden, and sad. When they are manic, they are either *happy or irritable*. About 1% of all adults in the United States suffer from bipolar disorder at some time during their life (Robins et al., 1984).

People in a manic phase have trouble inhibiting their impulses. Mental hospitals cannot install fire alarms in certain wards because manic patients will pull the alarm repeatedly. They make costly errors of judgment, such as investing large sums of money in highly risky or poorly considered ventures. Even after their friends warn them of the risks, they plunge on ahead.

The rambling speech of a manic person has been described as a "flight of ideas." The person starts talking about one topic, which suggests another, which in turn suggests another. Here is a quote from a manic patient:

> I like playing pool a lot, that's one of my releases, that I play pool a lot. Oh, what else? Bartend, bartend on the side, it's kind of fun to, if you're a bartender you can, you can see how people reacted, amounts of alcohol and different guys around, different chicks around, and different situations, if it's snowing outside, if it's cold outside, the weather conditions, all types of different types of environments and types of different types of people you'll usually find in a bar. (Hoffman, Stopek, & Andreasen, 1986, p. 835)

Some people experience a mild degree of mania ("hypomania") almost always. These people are productive, popular, extroverted, life-of-the-party types. Mania can become so serious, however, that it makes normal life impossible. The theatrical director Joshua Logan has described his own experiences with depression and mania. A few excerpts follow.

### A Self-Report: Depressive Phase

> I had no faith in the work I was doing or the people I was working with. . . . It was a great burden to get up in the morning and I couldn't wait to go to bed at night, even though I started not sleeping well. . . . I thought I was well but feeling low because of a hidden personal discouragement of some sort—something I couldn't quite put my finger on. . . . I just forced myself to live through a dreary, hopeless existence that lasted for months on end. . . .
>
> My depressions actually began around the age of thirty-two. I remember I was working on a play, and I was forcing myself to work. . . . I can remember that I sat in some sort of aggravated agony as it was read aloud for the first time by the cast. It sounded so awful that I didn't want to direct it. I didn't even want to see it. I remember feeling so depressed that I wished that I were dead without having to go through the shame and defeat of suicide. I couldn't sleep well at all, and sleep meant, for me, oblivion, and that's what I longed for and couldn't get. I didn't know what to do and I felt very, very lost. (Fieve, 1975, pp. 42–43)

### A Self-Report: Manic Phase

Here, Logan describes his manic experiences:

> Finally, as time passed, the depression gradually wore off and turned into something else, which I didn't understand either. But it was a much pleasanter thing to go through, at least at first. Instead of hating everything, I started liking things—liking them too much, perhaps. . . . I put out a thousand ideas a minute: things to do, plays to write, plots to write stories about. . . .
>
> I decided to get married on the spur of the moment. . . . I practically forced her to say yes. Suddenly we had a loveless marriage and that had to be broken up overnight. . . .
>
> I can only remember that I worked constantly, day and night, never even seeming to need more than a few hours of sleep. I always had a new idea or another conference. . . . It was an exhilarating time for me.
>
> It finally went too far. In the end I went over the bounds of reality, or law and order, so to say. I don't mean that I committed any crimes, but I could easily have done so if anyone had crossed me. I flew into rages if contradicted. I began to be irritable with everyone. Should a man, friend or foe, object to anything I did or said, it was quite possible that I could poke him in the jaw. I was eventually persuaded by the doctors that I was desperately ill and should go into the hospital. But it was not, even then, convincing to me that I was ill.
>
> There I was, on the sixth floor of a New York building that had special iron bars around it and an iron gate that had slid into place and locked me away from the rest of the world. . . . I looked about and saw that there was an open window. I leaped up on the sill and climbed out of the window on the ledge on the sixth floor and said, "Unless you open the door, I'm going to climb down the outside of this building." At the time, I remember feeling so powerful that I might actually be able to scale the building. . . . They immediately opened the steel door, and I climbed back in. That's where manic elation can take you. (Fieve, 1975, pp. 43–45)

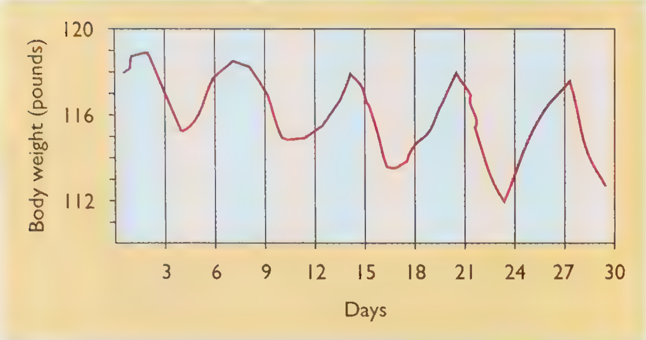


**FIGURE 16.20** Records for a man who had 3-day manic periods (pink) alternating with 3-day depressed periods (blue): Note that he lost weight during the manic times because of his high activity level. (Based on Crammer, 1959.)

## *Bipolar Cycles*

In most cases, periods of depression lasting months at a time alternate with somewhat shorter periods of mania. Less commonly, the depressed and manic phases may be very brief. Figure 16.20 shows the mood and body weight fluctuations for a manic-depressive man who had 3-day manic periods and 3-day depressed periods (Crammer, 1959).

Many artists, writers, and musical composers have suffered from either depression or bipolar disorder (Jamison, 1989). To test whether creative skills increase or decrease over various phases of the bipolar cycle, Robert Weisberg (1994) examined the works of the classical composer Robert Schumann, who is known to have had bipolar disorder. He found that Schumann produced more works during his manic phases than during his depressed phases. However, the compositions written during his depressed phases have been performed and recorded just as often as those written during his manic phases, on average. That is, the works that he composed during his depressed phases have been about as popular as those during his manic phases.

**CONCEPT CHECK**

**11.** What are the similarities and differences between seasonal affective disorder and bipolar disorder? (Check your answers on page 625.)

## *Lithium Therapy for Bipolar Disorder*

Many years ago, researcher J. F. Cade believed that uric acid might be effective for treating mania. To get uric acid to dissolve in water, he mixed it with lithium salts. The resulting mixture proved effective, but researchers eventually discovered that the benefits depended on the lithium salts, not the uric acid.

Lithium salts were soon adopted for use in the Scandinavian countries, but they were slow to be accepted in the United States. One reason was that drug manufacturers had no interest in marketing lithium pills. (Lithium is a natural substance and cannot be patented.) A second reason was that the lithium dosage must be carefully monitored. If the dosage is too low, it has no effect. If the dose is slightly higher than the recommended level, it produces nausea and blurred vision.

Properly regulated dosages of lithium are now a common and effective treatment for bipolar disorder (manic-depressive illness). Lithium reduces mania and protects the patient from relapsing into either mania or depression. It does not provide a permanent cure, so the patient must continue to take lithium pills every day. However, people can take the recommended doses of lithium for years without suffering unfavorable consequences (Schou, 1997). At this point, no one is certain how lithium relieves bipolar disorder. Many researchers believe that it works primarily on chemical pathways within the neurons, not on transmission at a particular kind of synapse (Manji, Potter, & Lenox, 1995). The other kind of medication often used for bipolar disorder is anticonvulsant drugs, such as valproate.

# *Suicide*

Most severely depressed people consider suicide, and many attempt it. Suicides also occur for other reasons, though. Some people commit suicide because of feelings of guilt or disgrace. Some commit suicide because a cult leader assures them that death is the route to salvation (Maris, 1997). Others have a painful, terminal illness and wish to hasten the end, sometimes with a physician's assistance. The causes of a suicidal wish are often unclear; for example, someone who requests a physician's assistance in dying may be suffering from treatable depression or insufficiently controlled pain (Farberman, 1997; Muskin, 1998).

Records on suicide cannot be fully accurate, because some people disguise their suicides to look like accidents,

Suicide can occur for a great many reasons. The Heaven's Gate cult in San Diego committed mass suicide because their leader assured them that death was the route to being rescued from Earth by an alien spacecraft.

and because many physicians, when in doubt, record the cause of death as something other than suicide. Nevertheless, Figure 16.21 shows the best evidence available for three countries in 1988 (Lester, 1996). Note the major differences in the suicide rate as a function of age, country, and gender. In almost every country, women make more suicide attempts than men, but more men than women die by suicide (Canetto & Sakinofsky, 1998; Cross & Hirschfeld, 1986). Most men who attempt suicide use guns or other violent means. Women are more likely to try poison, drugs, or other relatively slow, nonviolent methods that are less certain to be fatal (Rich, Ricketts, Fowler, & Young, 1988). Many people, especially women, who injure themselves in suicide attempts are believed to be crying out for help and not really intending to kill themselves. Unfortunately, some of these people actually die, and others become disabled for life.

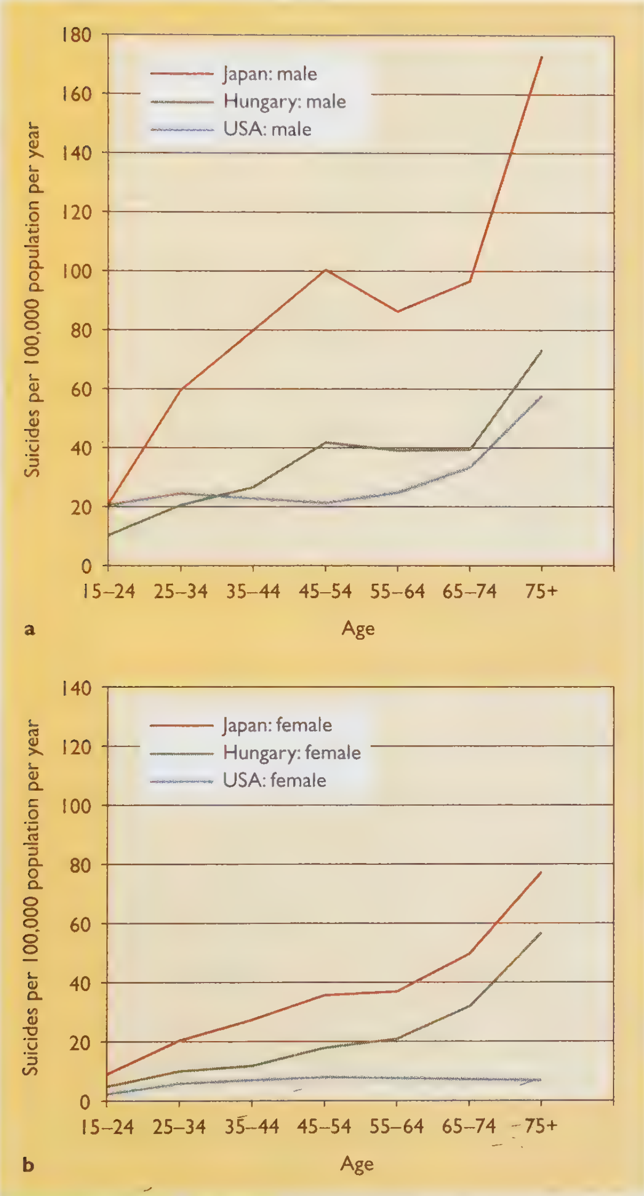


**FIGURE 16.21** Suicide rates differ as a function of age, gender, and culture. The rates shown here are for 1988; the rate has dropped since then for Hungary, presumably because of economic and social changes within the country. (Based on data of Lester, 1996.)

**TABLE 16.5 People Most Likely to Attempt Suicide**

- Depressed people, especially those with feelings of hopelessness (Beck, Steer, Beck, & Newman, 1993)
- People who have made previous suicide attempts (Beck, Steer, & Brown, 1993)
- People who have untreated psychological disorders (Brent et al., 1988), especially drug or alcohol abuse (Beck & Steer, 1989)
- People who have recently suffered the death of a spouse and men who have recently been divorced or separated, especially those who have little social support from friends and family (Blumenthal & Kupfer, 1986)
- People who, during their childhood or adolescence, lost a parent through death or divorce (Adam, 1986)
- People with guns in their home, particularly those with a history of violent attacks on others (Brent et al., 1988)
- People whose relatives have committed suicide (Blumenthal & Kupfer, 1986)
- People with low activity of the neurotransmitter serotonin in the brain (Roy, DeJong, & Linnoila, 1989)

Suicide follows no dependable pattern. Many people who attempt suicide give warning signals well in advance, but some do not. One study found that more than half of the people who made a serious suicide attempt decided on suicide less than 24 hours before making the attempt (Peterson, Peterson, O'Shanick, & Swann, 1985). However, certain factors are associated with an increased probability of attempting suicide. Anyone working with troubled people should be aware of these warning signals. Suicide attempts are most common among the types of people in Table 16.5. The same patterns have been found in the United States and China, so these are apparently not culture-specific (Cheng, 1995).

If you suspect that someone you know is thinking about suicide, what should you do? Treat the person like a normal human being. Don't assume that the person is so fragile that one wrong word will be disastrous. Don't be afraid to ask someone whether he or she has been contemplating suicide. The person may be relieved to find that you are not frightened by the thought and that you are willing to discuss it.

Most people who threaten suicide are crying out for help; they are feeling pain, either mental or physical. You may not be able to guess what kind of pain someone is feeling. Be prepared to listen.

Urge the person to get professional help. Most large cities have a suicide prevention hotline listed in the white pages of the telephone directory.

THE MESSAGE
## Normal Sadness Versus Major Depression

The capacity to feel sad is important. If you never felt sad, regardless of what misfortunes befell you, you might make disastrous decisions. Major depression is much more than just sadness; a severely depressed person loses the ability to feel happy, regardless of what good events do occur. Depression can be like a monster, so we need a wide arsenal of weapons to fight it.

## SUMMARY

* *Symptoms of depression.* A depressed person finds little interest or pleasure in life, feels worthless, powerless, and guilty, and may consider suicide. Such a person has trouble sleeping, loses interest in sex and food, and cannot concentrate. (page 614)
* *Predispositions.* Some people are predisposed to depression by genetic factors, by early experiences such as the loss of a parent, or by poor social support in adulthood. (page 615)
* *Sex differences.* Psychologists cannot convincingly explain why more women than men suffer depression. One hypothesis is that women are more likely to ruminate about their depression, and therefore aggravate and prolong it, whereas men are more likely to find some way of distracting themselves from their depression. (page 615)
* *Cognitive factors in depression.* People with a pessimistic explanatory style tend to blame themselves for their failures more than the evidence actually warrants. Depressed people almost invariably have an extremely pessimistic explanatory style, seeing evidence of their own failures in almost everything that happens. (page 616)
* *Cognitive therapy.* A frequently effective form of psychotherapy is to help depressed people reinterpret their experiences in a more favorable manner. (page 618)
* *Effects of antidepressant drugs.* Tricyclic drugs and MAOIs are used to treat depression. Both types of drugs prolong the stimulation of synaptic receptors by dopamine, norepinephrine, and serotonin. Second-generation antidepressants relieve depression by prolonging the action of serotonin only. With most people, second-generation antidepressants produce milder side effects. (page 618)
* *Advantages and disadvantages.* Psychotherapy is more likely to produce long-lasting benefits, but antidepressant drugs remain popular because of their convenience and lower cost. (page 619)
* *Electroconvulsive therapy.* Electroconvulsive therapy (ECT) has a long history of abuse; in modified form, ECT has made a comeback and is now helpful to some depressed people who fail to respond to antidepressant drugs. (page 620)
* *Seasonal affective disorder.* Seasonal affective disorder is an uncommon condition in which people become depressed during one season of the year. (page 620)
* *Bipolar disorder.* People with bipolar disorder alternate between periods of depression and periods of mania, when they engage in constant, driven, uninhibited activity. (page 621)
* *Lithium treatment.* Lithium salts are an effective treatment for bipolar disorder. (page 622)
* *Suicide.* Although it is difficult to know who will or will not attempt suicide, suicidal attempts are common among depressed people and people who show certain other warning signs. (page 623)

## Suggestions for Further Reading

Beers, C. W. (1948). *A mind that found itself.* Garden City, NY: Doubleday. (Original work published in 1908.) An autobiography of a man who recovered from a severe case of bipolar disorder.

Whybrow, P. C. (1997). *A mood apart.* New York: HarperCollins. A nontechnical description of the various forms of depression and their treatment.

## Terms

**major depression** a condition lasting most of the day, day after day, with a loss of interest and pleasure and a lack of productive activity (page 614)

**bipolar disorder** a condition in which a person alternates between periods of depression and periods of mania (page 614)

**postpartum depression** a period of depression that some women experience shortly after giving birth (page 615)

**learned-helplessness theory** the theory that some people become depressed because they have learned that they have no control over the major events in their lives (page 616)

**explanatory style** a tendency to accept one kind of explanation for success or failure more often than others (page 617)

**tricyclic drugs** drugs that block the reabsorption of the neurotransmitters dopamine, norepinephrine, and serotonin, after they are released by the terminal button, thus prolonging the effect of these neurotransmitters on the receptors of the postsynaptic cell (page 617)

**monoamine oxidase inhibitors (MAOIs)** drugs that block the metabolic breakdown of released dopamine, norepinephrine, and serotonin, thus prolonging the effects of these neurotransmitters on the receptors of the postsynaptic cell (page 618)

**second-generation antidepressants** drugs that block the reuptake of the neurotransmitter serotonin by the terminal button (page 618)

**electroconvulsive therapy (ECT)** a treatment using a brief electrical shock that is administered across the patient's head to induce a convulsion similar to epilepsy, sometimes used as a treatment for certain types of depression (page 620)

**seasonal affective disorder** a condition in which people become seriously depressed in one season of the year, such as winter (page 620)

**mania** a condition in which people are constantly active, uninhibited, and either happy or irritable (page 621)

## Answers to Concept Checks

9. Depressed people are less likely than others to buy a lottery ticket because they regard their chances of success as remote on any task. (page 618)
10. Mianserin should relieve depression and has in fact been used as an antidepressant. Although it acts by a different route from that of the tricyclics and MAOIs, it prolongs the stimulation of dopamine, norepinephrine, and serotonin receptors. (page 619)
11. Both seasonal affective disorder and bipolar disorder have repetitive cycles, sometimes with clocklike accuracy. However, people with bipolar disorder swing back and forth between two extremes, depression and mania, whereas most people with seasonal affective disorder alternate between depression and normal mood. (Some experience a slightly manic phase during the season opposite to the time of their depression.) (page 622)

## Web Resources

**Depression**
www.psych.org/public_info/depression.html
Manic-Depressive/Bipolar Disorder
www.psych.org/public_info/manic.html
The *Let's Talk Facts* series offers information about depression and manic-depressive/bipolar disorder, describing the disorders, the major symptoms, what we know about causes, and treatment.

MODULE 16.4

# Schizophrenia

What is schizophrenia?

What causes it?

What can be done about it?

How would you like to live in a world all your own? You can be the supreme ruler, and no one will ever criticize you or tell you what to do. You can tell other people—and even inanimate objects—what to do, and they will immediately obey. Each of your fantasies becomes a reality.

Perhaps that world sounds like heaven to you; I suspect it soon would be more like hell. Most of us enjoy the give and take of our interactions with other people; we enjoy struggling to achieve our fantasies more than we would enjoy their immediate fulfillment.

Some people with schizophrenia live practically in a world of their own, confusing fantasy with reality. They have trouble understanding what others say and difficulty making themselves understood. Eventually, they may retreat into a private existence and pay little attention to others.

## The Symptoms of Schizophrenia

The widely misunderstood term *schizophrenia* is based on Greek roots meaning "split mind." However, the term does *not* refer to a split into two minds or personalities. Many people use the term *schizophrenia* when they really mean *dissociative identity disorder* or *multiple personality*. People with dissociative identity disorder have several personalities, any one of which might be considered normal by itself.

In contrast, people suffering from schizophrenia have just one personality, but that personality is seriously disordered. The "split" in the schizophrenic "split mind" is a split between the intellectual and emotional aspects of the personality, as if the intellect and the emotions were no longer in contact with each other (Figure 16.22). A person suffering from schizophrenia may seem happy or sad without cause, may fail to show emotion in a situation that normally evokes them, or may even report bad news cheerfully or good news sadly.

To be diagnosed with **schizophrenia,** according to DSM-IV, *a person must exhibit a deterioration of daily activities, including work, social relations, and self-care. He or she must also exhibit at least two of the following: hallucinations, delusions, incoherent speech, grossly disorganized behavior, certain thought disorders, or a loss of normal emotional responses and social behaviors.* Exception: If someone's hallucinations or delusions are sufficiently severe, then no other symptoms are necessary. Finally, before assigning a diagnosis of schizophrenia, a psychologist or psychiatrist must rule out various other conditions that produce similar symptoms, including depression or bipolar illness, drug abuse, certain kinds of brain damage, the early stages of Huntington's disease, niacin deficiency, food allergies, and so forth.

About 1% of Americans are afflicted with schizophrenia at some point in life (Kendler, Gallagher, Abelson, & Kessler, 1996). Some sources cite somewhat higher or lower figures, depending on how many borderline cases they include. Schizophrenia occurs in all countries and in all ethnic groups, and is about equally common in men as in women.

Schizophrenia is most frequently diagnosed in young adults in their teens or 20s. Generally, men are diagnosed with schizophrenia earlier than women. A first diagnosis is rare after age 30 and unheard-of after 45. The onset can be sudden but is usually gradual. Most people with schizophrenia are described as having been "strange" children who had a short attention span, made few friends, often disrupted their classroom, and had mild thought disorders (Arboleda & Holzman, 1985; Parnas, Schulsinger, Schulsinger, Mednick, & Teasdale, 1982).

### SOMETHING TO THINK ABOUT

When we ask people to recall the childhood behavior of someone who later developed schizophrenia, what kinds of memory errors are likely, and why? (Recall the issues raised in Chapters 7 and 8.) ✷

### Hallucinations

**Hallucinations** are *sensory experiences that do not correspond to anything in the outside world.* Characteristically, people with schizophrenia hear voices and other sounds


**FIGURE 16.22** Although the term *schizophrenia* is derived from Greek roots meaning "split personality," it does not refer to cases where people alternate among different personalities. Rather, the term indicates a split between the intellectual and emotional aspects of a single personality.

that no one else hears. Not all schizophrenic people hear voices, but most people who do are suffering from schizophrenia. The voices may speak only nonsense, or else they may tell the person to carry out certain acts. Sometimes hallucinating people think the voices are real, sometimes they know the voices are coming from within their own head, and sometimes they are not sure (Junginger & Frame, 1985). Visual hallucinations are uncommon with schizophrenia, although some people have distorted or exaggerated visual experiences (Figure 16.23). Strong visual hallucinations are usually symptoms of drug abuse.

## *Delusions*

**Delusions** are *unfounded beliefs*. Three common types of delusions are delusions of persecution, grandeur, and reference: A **delusion of persecution** is *a belief that one is being persecuted*, that "people are out to get me." A **delusion of grandeur** is *a belief that one is unusually important*, perhaps a special messenger from God or a person of central importance to the future of the world. A **delusion of reference** is *a tendency to interpret all sorts of messages as if they were meant for oneself*. Someone with a delusion of reference may interpret a headline in the morning newspaper as a coded message or take a television announcer's comments as personal insults.

In some cases, it is fairly easy to identify a belief as delusional. For example, if someone who can hardly put together a complete sentence claims to be a messenger from the planet Zipton, chances are, the belief is a delusion. But what about someone who constantly sees evidence of government conspiracies in everyday events? Is that belief a delusion, or merely an unusual opinion? Members of religious and political minorities have many beliefs that other people regard as wrong, but unpopular views are not necessarily products of delusional thinking.

Furthermore, consider the following: During the Vietnam War, one U.S. Army unit massacred large numbers of unarmed women and children at My Lai. One of the soldiers, horrified at what the others were doing, refused to participate. After the war, he told a social worker about his recurring fears and nightmares stemming from that event. He also said that the other soldiers threatened to kill him if he told anyone what had happened and said that they might kill him anyway, just to prevent him from telling. When the social worker told her colleagues about this client, most of them labeled his story a delusion and suggested a diagnosis of schizophrenia. The story, however, was true (Scott, 1990).

In short: One should be cautious about labeling any belief a delusion and about diagnosing anyone as schizophrenic when the only symptom is an apparently delusional belief.

## *Disorders of Emotion and Movement*

Many people with schizophrenia show little sign of emotion. Their faces seldom express emotion, and they speak without the inflections most people use for emphasis. When they do show emotions, the expressions are inappropriate, such as laughing without apparent reason.

Some people with schizophrenia have a movement disorder called catatonia. **Catatonia** can take the form of either *rigid inactivity or excessive activity*; in either case, the person's movement pattern seems to be unrelated to events in the outside world.

**FIGURE 16.23** These portraits graphically illustrate the artist's progressive psychological deterioration. When well-known animal artist Louis Wain (1860–1939) began suffering delusions of persecution, his drawings showed a schizophrenic's disturbing distortions in perception.

## *The Thought Disorder of Schizophrenia*

One characteristic of schizophrenic thought is the use of loose and idiosyncratic associations, somewhat like the illogical leaps that occur in dreams. For example, one man used the words *Jesus, cigar,* and *sex* as synonyms. When he was asked to explain, he said they were all the same because Jesus has a halo around his head, a cigar has a band around it, and during sex people put their arms around each other.

Another characteristic of schizophrenic thought is difficulty using abstract concepts. For instance, many people with schizophrenia have trouble sorting objects into categories. Many also give strictly literal responses when asked to interpret the meaning of proverbs. Here are some examples (Krueger, 1978, pp. 196–197):

*Proverb:* People who live in glass houses shouldn't throw stones.
*Interpretation:* "It would break the glass."

*Proverb:* All that glitters is not gold.
*Interpretation:* "It might be brass."
*Proverb:* A stitch in time saves nine.
*Interpretation:* "If you take one stitch for a small tear now, it will save nine later."

People with schizophrenic thought disorder often misunderstand simple statements, because of their tendency to interpret everything literally. Upon being taken to the admitting office of a hospital, one person said, "Oh, is this where people go to admit their faults?"

Many schizophrenic people use vague, roundabout ways of saying something simple. For instance, one such person said, "I was born with a male sense" instead of "I'm a man." They often use many words to say nothing of importance, as in this excerpt from a letter one man wrote to his mother:

> I am writing on paper. The pen which I am using is from a factory called "Perry & Co." This factory is in England. I assume this. Behind the name of Perry Co. the city of London is inscribed; but not the city. The city of London is in England. I know this from my school-days. Then, I always liked geography. My last teacher in that subject was Professor August A. He was a man with black eyes. I also like black eyes. There are also blue and gray eyes and other sorts, too. I have heard it said that snakes have green eyes. All people have eyes. There are some, too, who are blind. These blind people are led about by a boy. It must be very terrible not to be able to see. There are people who can't see and, in addition, can't hear. I know some who hear too much. (Bleuler, 1911/1950, p. 17)

## The Distinction Between Positive and Negative Symptoms

Are all the various symptoms of schizophrenia independent, or do they form clusters? Many investigators distinguish between positive symptoms and negative symptoms of schizophrenia. (In this case, *positive* means *present* and *negative* means *absent;* they do not mean *good* and *bad.*) **Positive symptoms** are *behaviors that are notable because of their presence* in schizophrenia—symptoms such as hallucinations, delusions, and thought disorder. **Negative symptoms** are *behaviors that are notable by their absence.* For example, many people with schizophrenia show a lack of emotional expression, a lack of social interaction, a deficit of speech, a lack of ability to feel pleasure, and a general inability to take care of themselves.

The research supports a further distinction between two groups of positive symptoms—"positive psychotic" symptoms (hallucinations and delusions) and "positive disorganized" symptoms (thought disorder, bizarre behavior, and inappropriate emotions). Hallucinations and delusions form a natural cluster; most people who have either hallucinations or delusions have both. Similarly, the positive disorganized symptoms correlate with one another; people who have one of them tend to have the others also (Andreasen, Arndt, Alliger, Miller, & Flaum, 1995). The negative symptoms form a third cluster.

As a rule, both the positive psychotic and the positive disorganized symptoms fluctuate from time to time. Negative symptoms tend to be more consistent over time and more difficult to treat (Arndt, Andreasen, Flaum, Miller, & Nopoulos, 1995). People with many negative symptoms have an earlier onset of the disorder and worse performance in school and on the job (Andreasen, Flaum, Swayze, Tyrrell, & Arndt, 1990). These people are also more likely to show signs of brain abnormalities (Palmer et al., 1997).

# Types of Schizophrenia

People with schizophrenia vary considerably in their symptoms; they also vary in their family histories, their results on brain scans, and their response to treatment. What we call "schizophrenia" may in fact include two or more separate conditions that produce overlapping symptoms, as if medical doctors had failed to distinguish among migraine headaches, tension headaches, and brain tumors (Heinrichs, 1993). Currently, psychologists distinguish among four types of schizophrenia, based on the behavioral symptoms. These distinctions are useful for descriptive purposes, although they do not identify the underlying causes.

**Undifferentiated schizophrenia** is characterized by the *basic symptoms—deterioration of daily functioning plus some combination of hallucinations, delusions, inappropriate emotions, thought disorders, and so forth.* However, none of these symptoms is unusually pronounced or bizarre.

**Catatonic schizophrenia** is characterized by the *basic symptoms plus prominent movement disorders.* The affected person may go through periods of extremely rapid, mostly repetitive activity alternating with periods of total inactivity. During the inactive periods, he or she may hold a given posture without moving and may resist attempts to alter that posture (Figure 16.24). Catatonic schizophrenia is rare.

**Disorganized schizophrenia** is characterized by *incoherent speech, extreme lack of social relationships, and "silly" or odd behavior.* For example, one man gift wrapped one of his bowel movements and proudly presented it to his therapist. Here is a conversation with someone suffering from disorganized schizophrenia (Duke & Nowicki, 1979):

**Interviewer:** How does it feel to have your problems?
**Patient:** Who can tell me the name of my song? I don't know, but it won't be long. It won't be short, tall, none at all. My head hurts, my knees hurt—my nephew, his uncle, my aunt. My God, I'm happy . . . not a care in the world. My hair's been curled, the flag's unfurled. This is my country, land that I love, this is the country, land that I love.
**Interviewer:** How do you feel?
**Patient:** Happy! Don't you hear me? Why do you talk to me? (barks like a dog). (Duke & Nowicki, 1979, p. 162)

**FIGURE 16.24** A person suffering from catatonic schizophrenia can hold a bizarre posture for hours and alternate this rigid stupor with equally purposeless, excited activity. Such people may stubbornly resist attempts to change their behavior, but they need supervision to avoid hurting themselves or others. Catatonic schizophrenia is uncommon.

**Paranoid schizophrenia** is characterized by the *basic symptoms plus strong or elaborate hallucinations and delusions, especially delusions of persecution and delusions of grandeur.* Many people with paranoid schizophrenia are more cognitively intact than most people with other forms of schizophrenia. They can generally take care of themselves well enough to get through the activities of the day, except for their constant suspicions that "I am surrounded by spies" or that "evil forces are trying to control my mind."

Paranoid schizophrenia tends to run in different families from other types of schizophrenia (Farmer, McGuffin, & Gottesman, 1987). It usually develops gradually, somewhat later in life than other types of schizophrenia.

Many people fall on the borderline between two or more types of schizophrenia, perhaps switching back and forth between them. Switching is especially common between undifferentiated schizophrenia and one of the other types (Kendler, Gruenberg, & Tsuang, 1985).

## CONCEPT CHECK

12. Why are people more likely to switch between undifferentiated schizophrenia and one of the other types than, say, between disorganized schizophrenia and one of the other types? (Check your answer on page 635.)

# Causes of Schizophrenia

From all indications, schizophrenia has multiple causes. In most cases, it apparently begins with a genetic or other biological predisposition, which is then aggravated by stressful experiences. We shall examine what researchers currently understand about the causes of schizophrenia, but bear in mind that our understanding is tentative.

## Brain Damage

Unlike people suffering from most other psychological disorders, people suffering from schizophrenia show minor but widespread brain damage (see Figure 16.25). The cerebral cortex is somewhat shrunken in one-fourth to one-third of all schizophrenic patients, and the cerebral ventricles (fluid-filled spaces in the brain) are enlarged, on the average (Zipursky, Lim, Sullivan, Brown, & Pfefferbaum, 1992). Enlargement of the cerebral ventricles implies decreased space for neurons; thus, it suggests mild brain damage. Most people with schizophrenia have cognitive and memory deficits similar to those of people with damage to the prefrontal cortex (Seidman et al., 1995).

At a microscopic level, researchers have found smaller than normal neurons (Selemon, Rajkowska, & Goldman-Rakic, 1995), decreased metabolic activity in the brain (Berman, Torrey, Daniel, & Weinberger, 1992), abnormal locations of certain neurons in the brain (Akbarian et al., 1993), and decreased release of neurotransmitters (Glantz & Lewis, 1997). Each of these differences is especially marked in the prefrontal cortex. Indications of brain abnormalities have been reported in patients in cultures as different as the United States and Nigeria (Ohaeri, Adeyinka, & Osuntokun, 1995).

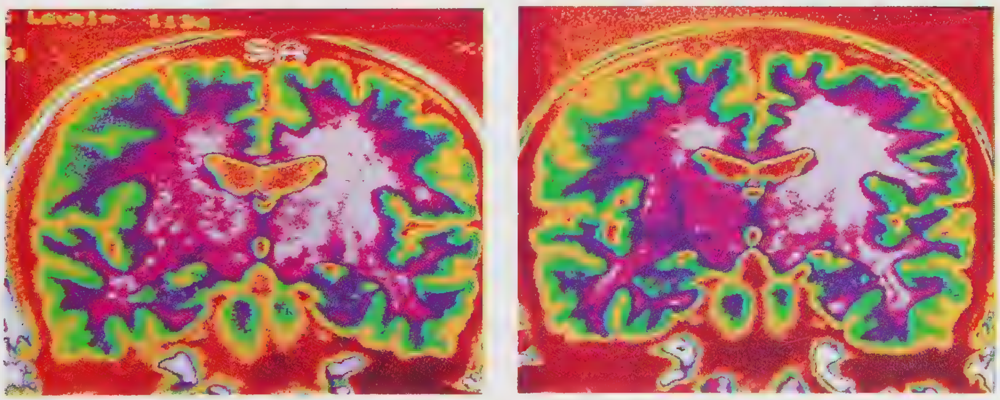

**FIGURE 16.25** Many (though not all) people with schizophrenia show signs of mild loss of neurons in the brain. Here, we see views of the brains of twins. The twin on the left has schizophrenia; the twin on the right does not. Note that the ventricles (near the center of each brain) are larger in the twin with schizophrenia. The ventricles are fluid-filled cavities; an enlargement of the ventricles implies a loss of brain tissue. (Photos courtesy of E. F. Torrey & M. F. Casanova / NIMH.)

Although the causes of these brain abnormalities are uncertain, mounting evidence suggests that the abnormalities develop early in life, either before birth or early after birth. People with schizophrenia do not show signs of increasing brain abnormality in adolescence or adulthood, though (Benes, 1995).

**CONCEPT CHECK**

**13.** Following a stroke, a patient shows symptoms similar to schizophrenia. Where is the brain damage probably located? (Check your answer on page 635.)

## *Genetics*

The evidence of a genetic basis rests primarily on studies of adopted children and comparisons of twins. With adopted children who eventually develop schizophrenia, schizophrenia is more common among their biological relatives than it is among their adoptive relatives (Kety et al., 1994). If one member of a pair of monozygotic (identical) twins develops schizophrenia, there is almost a 50% chance that the other will develop it too (Gottesman, 1991). (See Figure 16.26.) Furthermore, when one twin develops schizophrenia and the other does not, both twins run the same risk of passing schizophrenia on to their children (Gottesman & Bertelson, 1989; Kringlen & Cramer, 1989). Apparently, a gene or genes tends to increase the likelihood of schizophrenia. Even if a person with those genes does not actually develop schizophrenia, he or she still passes the genes on to the next generation.

Although nearly all researchers agree that genetic factors contribute to schizophrenia, our current research methods probably overestimate the role of genetics by underestimating the importance of the prenatal environment (Phelps, Davis, & Schartz, 1997). An adopted child shares not only the genes of his or her biological relatives, but also the prenatal environment provided by the biological mother. Most monozygotic twins share a single placenta before birth, whereas all dizygotic twins have separate placentas (Figure 16.27). Therefore, monozygotic twins have more similarities in prenatal influences than do dizygotic twins. Several types of evidence, discussed later, suggest that prenatal environment contributes to some cases of schizophrenia.

One focus of current research is to identify those people who have the presumed genes for schizophrenia but who do not show the symptoms. Identifying these people could be useful for better understanding the genetics and for understanding how genetic influences combine with environmental influences. Psychologists have examined young people who are related to people with schizophrenia. Although these young people have not yet developed schizophrenia themselves, many of them will probably develop it later. Thus, any unusual behaviors they now show are possible "markers" of vulnerability to schizophrenia—that is, people with these behaviors are more likely than others to develop schizophrenia.

Those markers include several symptoms of mild brain damage or abnormality (Cannon et al., 1994) and a failure to habituate normally to a repeated sound (Hollister, Mednick, Brennan, & Cannon, 1994). That is, many of these young people fail to filter out irrelevant information, thus responding to a repeated noise as much as they do to an unfamiliar noise.

Another possible "marker" is an impairment of pursuit eye movements, the movements necessary for keeping one's eyes focused on a moving object. Most people

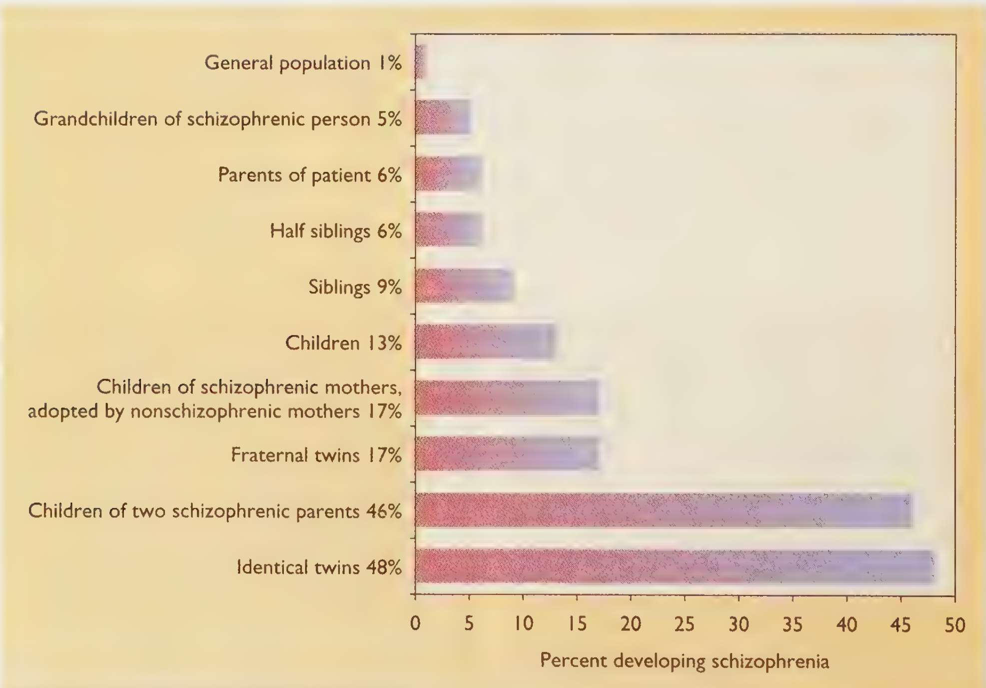


**FIGURE 16.26** The relatives of a schizophrenic person have an increased probability of developing schizophrenia themselves. Note that children of a schizophrenic mother have a 17% risk of schizophrenia even if adopted by a family with no schizophrenic members. (Based on data from Gottesman, 1991.)

with schizophrenia move their eyes in a series of rapid jerks instead of moving them smoothly (Sereno & Holzman, 1993); they show this same impairment before they develop schizophrenia, during a schizophrenic episode, and after successful therapy (Holzman, 1988). The same impairment also occurs in many of their close relatives but seldom in people who do not have a relative with schizophrenia (Keefe et al., 1997). Thus, this impaired eye movement may help us to identify people who have the genes for schizophrenia.

Schizophrenia is slightly more common for children born in winter than those born at other times. However, the critical factor is probably the weather during the second trimester of pregnancy. Those born in winter were in the second trimester during fall, the main time for viral and bacterial epidemics.

**SOMETHING TO THINK ABOUT**

People with schizophrenia, especially men, are less likely than others to have children. So, it is difficult to imagine how a gene that leads to schizophrenia could spread enough to affect 1% of the population. Can you imagine a possible explanation? ✸

## The Neurodevelopmental Hypothesis

Although schizophrenia most often occurs in people who have relatives with schizophrenia, it also occurs in those who do not. Most researchers now accept the **neurodevelopmental hypothesis,** *the idea that schizophrenia originates with impaired development of the nervous system before or around the time of birth, possibly but not necessarily for genetic reasons.* For example, the probability of schizophrenia is higher than normal following these events relating to early brain development:

- The patient's mother had a very difficulty pregnancy, labor, or delivery (Hultman, Öhman, Cnattingius, Wieselgren, & Lindström, 1997).
- The mother was near starvation during the early stages of pregnancy (Susser et al., 1996).
- The mother had an Rh-negative blood type and her baby was Rh-positive. The risk of schizophrenia is especially strong for her second and later Rh-positive children, especially boys (Hollister, Laing, & Mednick, 1996).

Furthermore, *a person born in the winter months is slightly more likely to develop schizophrenia than a person born at any other time* (Bradbury & Miller, 1985). No other psychological disorder has this characteristic. Moreover, investigators have clearly demonstrated this **season-of-birth effect** only in the northern climates, not near the equator. Evidently, something about the weather near the time of birth contributes to some people's vulnerability to schizophrenia.

One possible explanation relates to the fact that influenza and other epidemics are most common in the fall, especially in northern climates. If a woman catches influenza or a similar disease while she happens to be in the second trimester of pregnancy, her illness can impair critical stages of brain development occurring within her fetus at that time. The virus does not cross the placenta into the fetus (Taller et al., 1996), so the problem is not the virus itself but probably the mother's fever, which can impair development of fetal neurons (Laburn, 1996). According to several studies, in years having a major influenza epidemic in the fall, the babies born 3 months later (in the winter) have an increased risk for schizophrenia, as diagnosed 20 or more years later (Kendell & Kemp, 1989; Mednick, Machon, & Huttunen, 1990; Torrey, Rawlings, & Waldman, 1988).

a

b

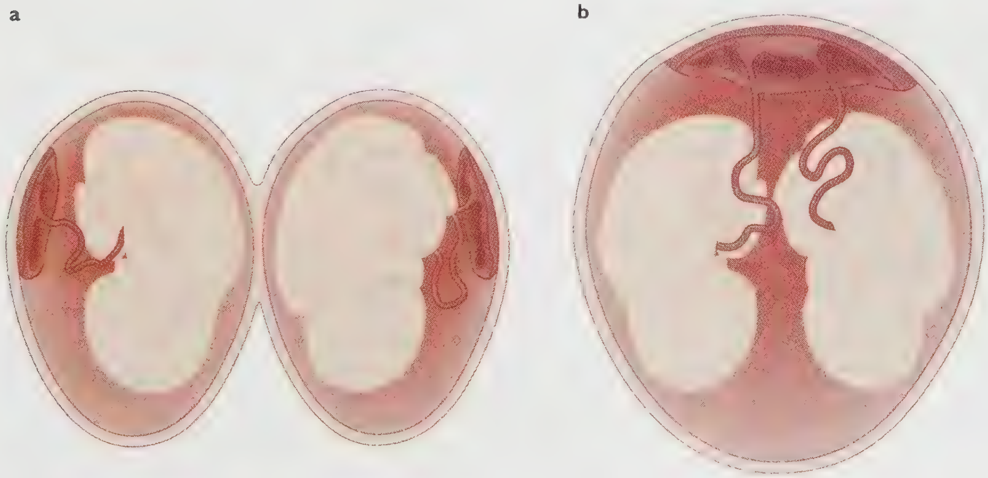

**FIGURE 16.27** All dizygotic (fraternal) twins develop with separate placentas and therefore separate blood supplies. Most monozygotic (identical) twins develop with a single placenta and therefore always have the same blood supply, the same hormone levels, and so forth. Some of the similarities seen between monozygotic twins may be due to their similar prenatal environments, not just to their genetic similarity.

You might ask, if the brain damage occurs before or near the time of birth, why do the symptoms emerge so much later? One possible answer is that certain parts of the brain, especially the prefrontal cortex, go through a critical stage of development during the second trimester of pregnancy but do not become fully functional until adolescence. As the brain begins to rely more and more on those areas, the effects of the brain damage become more evident (Weinberger, 1987).

### *Experience*

Assuming that genes or prenatal problems predispose certain people to schizophrenia, what environmental factors determine whether those genes will be expressed as schizophrenia? Decades ago, psychologists suggested that mothers who gave a confusing mixture of "come here" and "go away" signals were likely to induce schizophrenia in their children.

That theory is worth mentioning here only for the sake of explaining why it has been discarded. One reason is that it does not fit the data on adoptions. The child of a schizophrenic parent who is adopted by normal parents has a high risk of developing schizophrenia, whereas other adopted children reared in the same family generally develop normally. So, it seems unlikely that a mother's mixed signals cause schizophrenia.

Furthermore, the "bad mother" theory does not fit the course of the disorder. If the mother's behavior were the main cause of the problem, we would expect the child to improve after being separated from her. In fact, schizophrenia usually develops in early adulthood—when most people become independent of their parents.

Many researchers today attribute schizophrenia to a biological predisposition that is aggravated by stressful experiences (Walker & Diforio, 1997). As an analogy, someone with an injured foot may be able to walk normally most of the time but will start to limp when walking up a steep hill or when carrying a heavy backpack. The limp depends on both the foot injury (biological predisposition) and the hill or load (stress from the environment).

## *Drug Therapies for Schizophrenia*

Before the discovery of effective drugs to combat schizophrenia, the outlook for people with the disorder was bleak. Usually, people underwent a gradual deterioration, interrupted by periods of partial recovery. Many spent virtually their entire adult life in mental hospitals.

During the 1950s, researchers discovered the first effective antischizophrenic drug: chlorpromazine (klor-PRAHM-uh-ZEEN; trade name: Thorazine). *Drugs that relieve schizophrenia* are known as **antipsychotic drugs** or *neuroleptic drugs.* Chlorpromazine and other antipsychotic drugs, including haloperidol (HAHL-o-PAIR-ih-dol; trade name: Haldol), have enabled many schizophrenic people to escape lifelong confinement in a mental hospital. Although these drugs do not cure the disorder, a daily dosage does help to control it, much as daily insulin shots control diabetes. Since the 1950s, a majority of people with schizophrenia have improved enough to leave mental hospitals or to avoid ever entering one (Harding, Brooks, Ashikaga, Straus, and Breier, 1987).

All antipsychotic drugs share one characteristic: They block dopamine synapses in the brain. In fact, their therapeutic effectiveness is nearly proportional to their ability to block those synapses (Seeman & Lee, 1975). Furthermore, large doses of amphetamines, cocaine, or other drugs that stimulate dopamine activity can induce a temporary state that resembles schizophrenia. These phenomena have led to the **dopamine hypothesis of schizophrenia,** which holds that *the underlying cause of schizophrenia is excessive stimulation of certain types of dopamine synapses.*

Measurements from blood and other body fluids have generally found nearly normal levels of dopamine and its metabolic breakdown products, however (Jaskiw & Weinberger, 1992). According to an alternative view, the **glutamate hypothesis of schizophrenia,** *the underlying problem causing schizophrenia is deficient stimulation of certain glutamate synapses.* In many brain areas, dopamine inhibits glutamate activity, so drugs that block dopamine synapses would increase the activity of glutamate synapses. The glutamate hypothesis is supported by the fact that prolonged use of the drug *phencyclidine* ("angel dust"), which inhibits glutamate receptors, produces symptoms that match schizophrenia even more closely than do cocaine and amphetamines, which stimulate dopamine synapses (Olney & Farger, 1995).

Antipsychotic drugs take effect gradually and produce variable degrees of recovery. As a rule, antipsychotic drugs produce their clearest effects if treatment begins shortly after a sudden onset of schizophrenia. The greater someone's deterioration before drug treatment begins, the less the recovery will be. Most of the recovery that will ever take place emerges gradually during the first month (Szymanski, Simon, & Gutterman, 1983). Beyond that point, the drugs merely maintain behavior but do not improve it. When affected people stop taking the drugs, the symptoms return and, in most cases, worsen (Figure 16.28).

### *Side Effects of Drug Therapies for Schizophrenia*

Antipsychotic drugs produce unwelcome side effects in many people. The most serious, **tardive dyskinesia** (TAHRD-eev DIS-ki-NEE-zhuh), *a condition characterized by tremors and involuntary movements* (Chakos et al., 1996), develops gradually after years of taking antipsychotic drugs, especially with people who take large amounts of the drugs. Tardive dyskinesia is presumably related to activity at the dopamine

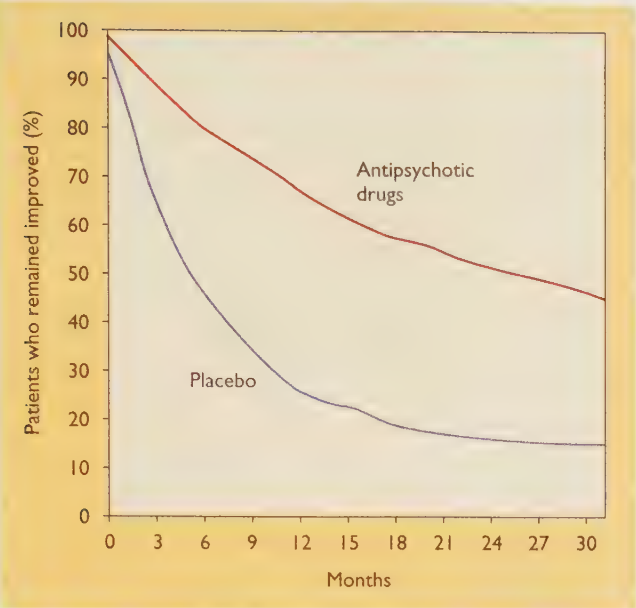


**FIGURE 16.28** This graph indicates that during 2½ years following apparent recovery from schizophrenia, the percentage of schizophrenic patients who remained "improved" is higher in the group that received continuing drug treatment than in the placebo group. But the graph also shows that antipsychotic drugs do not always prevent relapse. (Based on Baldessarini, 1984.)

synapses, some of which control movement; however, the exact explanation remains uncertain (Andersson et al., 1990).

Researchers have sought new drugs that can combat schizophrenia without causing tardive dyskinesia. On the theory that the symptoms of schizophrenia depend on one type of dopamine synapse and that tardive dyskinesia reflects changes at another type of dopamine synapse, researchers have experimented with drugs that specifically antagonize one or another type of dopamine receptor. Research has also focused on the relatively new drugs *clozapine* and *risperidone*, which relieve schizophrenia by combining moderate effects on dopamine synapses with additional effects on serotonin synapses. These new drugs have shown significant promise for relieving schizophrenia with a minimum risk of tardive dyskinesia; they also tend to relieve the negative symptoms of schizophrenia (such as social withdrawal) that other antipsychotic drugs fail to address (Carpenter, 1995; Meltzer, 1995). Unfortunately, clozapine produces serious side effects of its own, including an impairment of the immune system.

## CONCEPT CHECK

**14.** Why is early diagnosis of schizophrenia important? (Check your answer on page 635.)

## *Family Therapy for Schizophrenia*

The degree of recovery produced by antipsychotic drugs varies. Even someone who responds well to the drugs and appears to be living fairly normally can have a sudden relapse of symptoms. Much of this fluctuation in outcome reflects stressors in the person's family environment.

If you had a brother, sister, son, or daughter with schizophrenia, how would you react? We would all like to think that we would be 100% supportive and sympathetic to this poor, troubled person. However, family members are human beings, too, and—after years of dealing with someone who says and does strange things and requires extensive attention just to get through the normal activities of the day—even the most saintly of relatives will occasionally lose patience and make *hostile or critical comments,* known as **expressed emotion.** Researchers have found that the more frequently a person is exposed to expressed emotion, the greater the probability of a relapse into severe schizophrenic symptoms (Butzlaff & Hooley, 1998). In several studies, family members were taught to reduce their expressed emotion; the result has been fewer relapses into schizophrenia, as compared to control groups where families received no such training (de Jesus Mari & Streiner, 1994).

The results regarding expressed emotion suggest one possible explanation for some cross-cultural differences: Compared to North America and western Europe, schizophrenia in India and the Arab countries tends to be less severe and marked by fewer relapses. In these cultures, troubled people are generally cared for by a large extended family including cousins, aunts and uncles, and so forth, and not just by the immediate family. By sharing care, these families decrease the amount of strain on any one individual. Relatives can thus maintain their patience and good spirits; they show much fewer expressed emotions than their U.S. counterparts (El-Islam, 1982; Leff et al., 1987; Wig et al., 1987).

## THE MESSAGE *The Elusiveness of Schizophrenia*

You could meet two people both diagnosed with schizophrenia who nevertheless had relatively little in common. One has hallucinations and delusions; the other has a thought disorder and a lack of emotional expression. One has relatives with schizophrenia; the other doesn't. One has evidence of brain abnormalities; the other doesn't. As you can imagine, it is difficult to draw generalizations that apply to all people with schizophrenia, so many researchers still wonder whether we are dealing with one disorder or several. We need more research to understand the disorder itself as well as its causes and treatment.

## SUMMARY

* *Symptoms of schizophrenia.* A person with schizophrenia is someone whose everyday functioning has deteriorated

over a period of at least 6 months and who shows at least two of the following symptoms: hallucinations (mostly auditory), delusions, weak or inappropriate emotional expression, catatonic movements, and thought disorder. (page 626)

* *Thought disorder of schizophrenia.* The thought disorder of schizophrenia is characterized by loose associations, impaired use of abstract concepts, and vague, wandering speech that conveys little information. (page 627)
* *Positive and negative symptoms.* Positive symptoms are behaviors that attract attention by their presence, such as hallucinations and delusions. Negative symptoms are behaviors that are noteworthy by their absence, such as impaired emotional expression. (page 628)
* *Types of schizophrenia.* What we call *schizophrenia* may in fact consist of two or more separate conditions with overlapping symptoms. Psychologists distinguish four types of schizophrenia: undifferentiated, catatonic, disorganized, and paranoid. Some authorities believe paranoid schizophrenia resembles depression more than it does other types of schizophrenia. (page 628)
* *Brain abnormalities.* Many people with schizophrenia show indications of mild brain abnormalities, especially in the prefrontal cortex. The abnormalities apparently develop early in life and do not grow worse over time. (page 629)
* *Genetic influences.* Much evidence indicates that it is possible to inherit a predisposition toward schizophrenia, although little is known about how the genes exert their influence. (page 630)
* *Neurodevelopmental hypothesis.* Many researchers believe that schizophrenia originates with abnormal brain development before or around the time of birth, sometimes for genetic reasons but sometimes for other reasons, such as a fever the mother had during pregnancy. (page 631)
* *Antipsychotic drugs.* Drugs that alleviate schizophrenia block dopamine synapses. Results are best if treatment begins before the person has suffered serious deterioration. (page 632)
* *Neurotransmitters.* The effectiveness of dopamine blockers in alleviating schizophrenia has suggested that the underlying problem might be excessive dopamine activity. However, people with schizophrenia appear to have normal dopamine levels. An alternative hypothesis is that the underlying problem is a deficiency of glutamate, a neurotransmitter inhibited by dopamine. (page 632)
* *Family therapy.* The family's hostile comments (expressed emotion) toward someone with schizophrenia increase the risk of renewed symptoms. Reducing the family's expressed emotion improves the patient's chances for lasting recovery. (page 633)

## Suggestions for Further Reading

Andreasen, N. C. (1994). *Schizophrenia: From mind to molecule.* Washington, DC: American Psychiatric Press. Review of schizophrenia by one of the leading researchers.

Heston, L. L. (1992). *Mending minds.* New York, W. H. Freeman. A nontechnical account of the use of drugs in psychiatry.

## Terms

**schizophrenia** a condition marked by deterioration of daily activities over a period of at least 6 months, plus hallucinations, delusions, flat or inappropriate emotions, certain movement disorders, or thought disorders (page 626)

**hallucinations** sensory experiences that do not correspond anything in the outside world (page 626)

**delusions** unfounded beliefs (page 627)

**delusion of persecution** the belief that one is being persecuted (page 627)

**delusion of grandeur** the belief that one is unusually important (page 627)

**delusion of reference** the tendency to interpret all sorts of messages as if they were meant for oneself (page 627)

**catatonia** a movement disorder, consisting of either rigid inactivity or excessive activity (page 627)

**positive symptoms** characteristics present in people with schizophrenia and absent in others—such as hallucinations, delusions, abnormal movements, and thought disorder (page 628)

**negative symptoms** symptoms that are present in other people—such as the ability to take care of themselves—but absent in schizophrenic people (page 628)

**undifferentiated schizophrenia** a type of schizophrenia characterized by the basic symptoms but no unusual or especially prominent symptoms (page 628)

**catatonic schizophrenia** a type of schizophrenia characterized by the basic symptoms plus prominent movement disorders (page 628)

**disorganized schizophrenia** a type of schizophrenia characterized by incoherent speech, extreme lack of social relationships, and "silly" or odd behavior (page 628)

**paranoid schizophrenia** a type of schizophrenia characterized by the basic symptoms plus strong or elaborate hallucinations and delusions (page 629)

**neurodevelopmental hypothesis** the idea that schizophrenia originates with impaired development of the nervous system before or around the time of birth, possibly but not necessarily for genetic reasons (page 631)

**season-of-birth effect** the tendency for people born in the winter months to be slightly more likely than other people are to develop schizophrenia (page 631)

**antipsychotic drugs** drugs that relieve schizophrenia (page 632)

**dopamine hypothesis of schizophrenia** the theory that the underlying cause of schizophrenia is excessive stimulation of certain types of dopamine synapses (page 632)

**glutamate hypothesis of schizophrenia** the view that the underlying problem causing schizophrenia is deficient stimulation of certain glutamate synapses (page 632)

**tardive dyskinesia** a disorder characterized by tremors and involuntary movements (page 632)

**expressed emotion** hostile or critical comments directed toward a person with a psychiatric disorder such as schizophrenia (page 633)

## Answers to Concept Checks

**12.** With any disorder, symptoms are more severe at some times than at others. Whenever any of the special symptoms of catatonic, disorganized, or paranoid schizophrenia become less severe, the person is left with undifferentiated schizophrenia. To shift between any two of the other types, a person would have to lose the symptoms of one type and gain the symptoms of the other type. (page 629)

**13.** The damage is probably located in the frontal or temporal lobes of the cerebral cortex, the area that is generally damaged in people with schizophrenia. (page 630)

**14.** Antipsychotics are more helpful to people in the early stages of schizophrenia than to those who have deteriorated severely. However, psychiatrists do not want to administer antipsychotics to people who do not need them because of the risk of tardive dyskinesia. Consequently, early and accurate diagnosis of schizophrenia is helpful. (page 633)

## Web Resources

**Schizophrenia**

www.psych.org/public_info/schizo.html

This *Let's Talk Facts* pamphlet covers the major symptoms of schizophrenia, what we know about its causes, and treatment. The American Psychiatric Association is also conducting a Schizophrenia Awareness Project (www.degnanco.com/schizophrenia/main.html) to help patients, spouses, and siblings understand the illness and its treatment, as well as how to help the patient on the road to recovery.

**Psychiatric Medication**

http://www.psych.org/public_info/medication.html

**Psychiatric Hospitalization**

http://www.psych.org/public_info/hospital.html

The *Let's Talk Facts* series offers information about psychiatric medication and hospitalization.